

Prime offsets around even squares: local restrictions and finite verification

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Abstract

For even n , we ask whether each adjacent square gap contains a prime whose distance from n^2 is also prime. We prove elementary restrictions, count admissible offset residues for a fixed center, and compare the two directions through a simultaneous local count. An exact search, independently checked by trial division, finds bounded witnesses in both directions for every even $2 \leq n \leq 10\,000$. Published minimal Goldbach-partition computations give a larger finite range on the lower side. The universal assertions remain conjectural.

1 The problem

Let $n \geq 2$ be even. An *upper witness* is a prime $q < 2n + 1$ for which $n^2 + q$ is prime. A *lower witness* is a prime $q < 2n - 1$ for which $n^2 - q$ is prime. The directions may use different offsets. Define the bounded minima

$$Q_+(n) = \min\{q : q \text{ prime, } q < 2n + 1, n^2 + q \text{ prime}\},$$
$$Q_-(n) = \min\{q : q \text{ prime, } q < 2n - 1, n^2 - q \text{ prime}\},$$

with value ∞ when the relevant set is empty. The upper and lower prime-offset conjectures assert, respectively, that these minima are finite for every even $n \geq 2$.

The strict bounds give

$$n^2 < n^2 + q < (n + 1)^2, \quad (n - 1)^2 < n^2 - q < n^2.$$

At $n = 2$, the minima are $Q_+(2) = 3$ and $Q_-(2) = 2$, giving primes 7 and 2. For every even $n \geq 4$, the offset 2 fails in both directions: $n^2 \pm 2$ is even and greater than 2.

Proposition 1. *If both prime-offset conjectures hold, every interval $(m^2, (m + 1)^2)$, $m \geq 1$ an integer, contains a prime.*

Proof. For even m , use an upper witness at $n = m$. For odd m , use a lower witness at the even center $n = m + 1$. The displayed strict inequalities place the prime in the required interval. The lower witness at $n = 2$ also handles $m = 1$. $\square$

This is a conditional implication for Legendre's conjecture. Prime existence in a square gap by itself imposes no primality condition on the displacement.

2 Elementary arithmetic restrictions

Recall why coprime integers a, b have integers x, y with $ax + by = 1$. Take the least positive integer d of the form $ax + by$. Division with remainder of a or b by d shows that any nonzero remainder would be a smaller positive combination. Thus d divides both a and b , so $d = 1$. This is Bézout's identity in the coprime case.

We use the elementary fact that a prime ℓ dividing n^2 divides n . Indeed, if $\ell \nmid n$, Bézout's identity gives an inverse of n modulo ℓ ; multiplying $n^2 \equiv 0$ by its square would give $1 \equiv 0$.

Proposition 2. *If $n \geq 4$ is even and a prime q divides n , then both $n^2 + q$ and $n^2 - q$ are composite.*

Proof. Both values are multiples of q . Since $q \leq n$, $n^2 - q \geq n^2 - n > n \geq q$. Thus each value has a proper prime divisor. $\square$

Corollary 3. *For every real $B \geq 2$, infinitely many even n have no successful prime offset $q \leq B$ in either direction.*

Proof. Let P be the product of all primes at most B . It includes 2, so all multiples $n = kP$ are even. For the infinitely many such $n \geq 4$, every prime $q \leq B$ divides n and Proposition 2 applies. $\square$

The statement also rules out every fixed finite collection of prime offsets: choose B at least as large as all its elements. It does not assert that a larger witness exists. If the two conjectures hold, both minima exceed B at these centers; without them, either minimum could be ∞ . For example, $n = 30$ eliminates 2, 3 and 5, but its actual minima are 7 and 13.

Proposition 4. *Let $n \geq 4$ be even with $3 \nmid n$, and let $q > 3$ be prime. Within the respective offset bounds, an upper witness requires $q \equiv 1 \pmod{3}$, and a lower witness requires $q \equiv 2 \pmod{3}$.*

Proof. Here $n^2 \equiv 1 \pmod{3}$, while q is congruent to 1 or 2. If $q \equiv 2$, then $3 \mid n^2 + q$. If $q \equiv 1$, then $3 \mid n^2 - q$. In the lower bounded interval $n^2 - q > (n - 1)^2 \geq 9$, and the upper value is larger still, so a value divisible by 3 is composite. $\square$

If $3 \mid n$, neither nonzero residue of q is excluded by this modulus. The offset $q = 3$ itself fails by Proposition 2. When $3 \nmid n$, $q = 3$ must instead be tested separately.

More generally, a congruence filter may reject a positive integer v when $\ell \mid v$ and $v > \ell$. It must retain $v = \ell$: divisibility by a small prime is not sufficient to reject that prime itself. This equality exception is essential for the lower witness at $n = 2$.

3 Counting local conditions at a fixed center

Fix an even n and a finite set S of distinct primes, with $M = \prod_{\ell \in S} \ell$. The empty product is 1. For $s \in \{1, -1\}$ set

$$A_s(n; M) = \{a \bmod M : \gcd(a, M) = \gcd(n^2 + sa, M) = 1\}.$$

The variable a ranges over all residue classes, not just primes. This distinction makes the following count exact.

Proposition 5. *For either sign,*

$$|A_s(n; M)| = \prod_{\ell \in S} (\ell - \nu_\ell(n)), \quad \nu_\ell(n) = \begin{cases} 1, & \ell \mid n, \\ 2, & \ell \nmid n. \end{cases}$$

Moreover, $a \mapsto -a$ is a bijection from $A_+(n; M)$ to $A_-(n; M)$.

Proof. At a single prime ℓ , exclude $a \equiv 0$ and $a \equiv -sn^2$. These residues coincide exactly when $\ell \mid n^2$, equivalently $\ell \mid n$. Thus there are $\ell - \nu_\ell(n)$ choices.

For completeness, the Chinese remainder theorem gives a bijection between residues modulo M and tuples of residues modulo the primes in S . Write $M_\ell = M/\ell$ and choose u_ℓ with $u_\ell M_\ell \equiv 1 \pmod{\ell}$, possible by Bézout's identity. The tuple (a_ℓ) is represented by $\sum_{\ell \in S} a_\ell u_\ell M_\ell \bmod M$. Two representatives of the same tuple differ by a multiple of every prime in S , hence of their product. This proves both existence and uniqueness, and the local choice counts therefore multiply.

Finally, negation preserves coprimality of a with M and changes $n^2 + a$ into $n^2 - (-a)$. Negation is its own inverse. $\square$

The exact fraction in a complete period is $|A_s|/M = \prod_{\ell \in S} (1 - \nu_\ell(n)/\ell)$. Upper and lower counts agree over a complete residue period even though their allowed classes can differ. This does not imply equal counts of prime witnesses in the two bounded intervals.

For an odd ℓ , the ratio to two independent divisibility tests is

$$c_\ell(n) = \frac{1 - \nu_\ell(n)/\ell}{(1 - 1/\ell)^2} = \begin{cases} \ell/(\ell - 1), & \ell \mid n, \\ \ell(\ell - 2)/(\ell - 1)^2, & \ell \nmid n. \end{cases}$$

At $\ell = 2$ the ratio is 2 because n is even. Thus $C_y(n) = 2 \prod_{3 \leq \ell \leq y, \ell \text{ prime}} c_\ell(n)$ is a finite correction factor for $y \geq 2$. Its algebra is exact; interpreting it as a correction to prime frequencies requires a heuristic model. The center is fixed throughout: we are counting offset residues, not solutions of quadratic congruences as n varies.

4 Simultaneous local conditions

The same offset can be tested on both sides. Put

$$B(n; M) = A_+(n; M) \cap A_-(n; M).$$

Proposition 6. *For even n ,*

$$|B(n; M)| = \prod_{\ell \in S} (\ell - \mu_\ell(n)), \quad \mu_\ell(n) = \begin{cases} 1, & \ell \mid n, \\ 3, & \ell \nmid n. \end{cases}$$

If $3 \in S$ and $3 \nmid n$, then $B(n; M)$ is empty.

Proof. The forbidden residues are $0, n^2, -n^2$. If $\ell \mid n$, all three coincide. Otherwise ℓ is odd, since n is even. Then n^2 and $-n^2$ are distinct nonzero residues: equality would imply $2n^2 \equiv 0 \pmod{\ell}$. There are consequently three exclusions. The CRT bijection proved above multiplies the local counts. For $\ell = 3 \nmid n$, the factor is $3 - 3 = 0$. $\square$

Corollary 7. *For even $n \geq 4$ with $3 \nmid n$, a prime q that is a bounded witness in both directions must be 3.*

Proof. The offset 2 fails by parity. Every prime $q > 3$ has a nonzero residue modulo 3, and Proposition 4 requires incompatible residues for the two directions. Only $q = 3$ remains. $\square$

The exception can occur: $4^2 - 3 = 13$ and $4^2 + 3 = 19$ are both prime. It is not counted by $B(n; M)$ when $3 \mid M$, because the offset itself equals a sieving prime. At $n = 6$, the common witness $q = 5$ gives 31 and 41. There is no contradiction with the two offset conjectures, which allow different witnesses on the two sides.

The following exact counts use $S = \{2, 3, 5, 7\}$ and $M = 210$. Each entry is checked by enumerating all 210 residues.

n	$ A_+ $	$ A_- $	$ B $
2	15	15	0
4	15	15	0
6	30	30	16
10	20	20	0
14	18	18	0
30	40	40	32
42	36	36	24
210	48	48	48

These are counts of residue classes. Passing the local tests is not a certificate of primality. Genuine prime witnesses equal to a sieving prime, or having a shifted value equal to one, are separate exceptions.

5 Exact bounded computation

Let N be the largest even integer at most the requested limit. The reference search marks primes by the Eratosthenes sieve through $T = (N + 1)^2$. It lists prime offsets $2 \leq q \leq 2N$ in increasing order. For each even $2 \leq n \leq N$ and each sign s , it tests only $q < 2n + s$, recording the first witness or a failure.

The sieve is exact: every composite $v \leq T$ has a prime divisor $p \leq \sqrt{v}$. Otherwise two nontrivial factors would both exceed $\sqrt{v}$. For the least prime divisor, $v \geq p^2$, so the sieve marks v among the multiples starting at p^2 . No prime is marked. The search is in range because the upper shifted value is below $(n + 1)^2 \leq T$, and the lower one is positive.

The optional filter uses only proper divisors from $\{2, 3, 5, 7\}$; it retains equality with the divisor. A second calculation uses trial division, without that filter or the sieve, to check that the recorded offset is prime, that its shifted value is prime, and that every smaller prime offset fails. For a reported failure it checks every allowed prime offset. Trial division is exact for the same square-root divisor reason.

All 5,000 even inputs through 10 000 have witnesses in both directions. The statistics below describe the minima, not all witnesses.

Statistic	Upper	Lower
Even inputs	5 000	5 000
Failures	0	0
Largest minimum	397	467
Center of largest minimum	6 046	8 968
Mean	27.8088	22.3494
Median	13	11
95th percentile	97	83

The percentile is the $\lceil 0.95k \rceil$ -th sorted successful minimum, where $k = 5\,000$. If a search has failures, they are listed explicitly and descriptive statistics use successful inputs only.

`results/offsets.csv` retains every input and both minima; an empty field denotes ∞ . `records.csv` contains successive strict records, `summary.json` includes the data hash, and `local_counts.csv` contains the residue table. The programs use only the Python standard library. Tests compare independent searches, exercise equality and boundary cases, and detect invalid or nonminimal reported witnesses. Tests supplement the proofs.

The sieve's main array has $(N + 1)^2 + 1$ bytes, about 100 MB at $N = 10\,000$, with additional temporary allocations. The program caps the limit at 20,000. This implementation and these results make no claim for a run to 10^8 ; a different algorithm and retained outputs would be needed.

6 A larger lower-side range from published computation

Oliveira e Silva, Herzog and Pardi [1] verified even Goldbach partitions through $4 \cdot 10^{18}$. Their $p(E)$ denotes the smallest prime summand in a partition of the even integer $E > 4$. Section 2.1 and Table 4, printed page 2043, report the successive records of $p(E)$ over this range. The last and largest record is 9781. Thus their exhaustive computation supplies, for each such E in range, primes r, p with

$$E = r + p, \quad r \leq 9781.$$

This is an input from a published finite computation, not a theorem about all even integers and not a computation repeated in this repository.

Proposition 8 (Using published computational input). *Together with the small-case calculation above, that input verifies the lower prime-offset assertion for every even $2 \leq n \leq 2 \cdot 10^9$.*

Proof. If $4892 \leq n \leq 2 \cdot 10^9$ is even, then $n^2 \leq 4 \cdot 10^{18}$. Apply the published input at $E = n^2$ and set $q = r$. Then $n^2 - q = p$ is prime and

$$q \leq 9781 < 9783 \leq 2n - 1.$$

The remaining even centers $2 \leq n \leq 4890$ are included in the repository's independently checked search through 10 000. □

The threshold is strict: $2(4891) - 1 = 9781$, so the published uniform bound alone would not suffice there. The first even center above this threshold is 4892. The even-square endpoint is exact: $(2 \cdot 10^9)^2 = 4 \cdot 10^{18}$.

This deduction needs the bound on the *minimal* smaller summand. A Goldbach representation with no bound on its summands would not imply $q < 2n - 1$. It also does not yield an upper witness: the upper question asks for the prime difference $p - q = n^2$, whereas the input gives the sum $p + q = n^2$. Nor does a prime in the upper square gap guarantee that its displacement from n^2 is prime.

Only two summary facts from the external computation are used: its verified endpoint and its largest minimal summand. No third-party dataset is redistributed or silently loaded by the programs. The source links and page reference are recorded in `results/README.md`.

The established ranges therefore have different provenance:

- Both sides through $n = 10\,000$: this repository's exact search and independent verification.
- Lower side through $n = 2 \cdot 10^9$: an elementary deduction using the published exhaustive computation and the repository's small cases.

No universal prime-existence statement follows from either finite range.

7 Heuristics, limitations and further questions

A rough model for prime offsets $q < H$ uses approximately $H/\log H$ prime candidates and a prime frequency near n^2 of $1/(2\log n)$. If H is small compared with n^2 , an uncorrected expected count is

$$\lambda_0(n, H) \approx \frac{H}{2\log n \log H}.$$

Parity doubles this baseline because odd q gives an odd shifted value. The finite factor $C_y(n)$ describes additional congruence corrections, but the exact residue counts do not prove that prime candidates in a short interval follow this model.

Solving the modeled count-of-order-one equation suggests a typical first-offset scale of order $\log n \log \log n$, up to local factors and constants. This is a heuristic interpretation, not a proved growth law. Records over many centers are a different statistic; the finite table gives no asymptotic bound for either minimum.

For H near $2n$, the model predicts a count of order $n/(\log n)^2$. A large expected count alone does not force existence. For example, a random variable equal to 0 or $2n$ with equal probabilities has expectation n and a fixed probability of zero. A formula such as $\Pr(\text{no witness}) \approx e^{-\lambda}$ requires an extra Poisson assumption. Even a decreasing modeled failure probability would not prove success for every even integer.

The proved results concern obstructions, exact finite residue counts, and a conditional implication between conjectures. The existence and minimum tables are finite computations. The larger lower range relies on external computational evidence. The two universal offset assertions, the converse of Proposition 1, and asymptotic laws for the minima are not established here.

Three further investigations follow naturally:

1. Group the finite minima by $n \bmod 30$ or divisibility by small primes. Compare means and quantiles separately from records.
2. Count common bounded witnesses, separating $3 \mid n$ from $3 \nmid n$ and retaining the exceptional offset 3.
3. For a fixed center and cutoff, compare actual witness counts with locally admissible candidates. Quantify the information lost when a finite residue filter is used as a model for primality.

None requires enlarging the project's prime-existence claims.

References

- [1] T. Oliveira e Silva, S. Herzog and S. Pardi, *Empirical verification of the even Goldbach conjecture and computation of prime gaps up to $4 \cdot 10^{18}$* , Mathematics of Computation **83** (2014), no. 288, 2033–2060. doi:10.1090/S0025-5718-2013-02787-1. Section 2.1, Table 4, p. 2043.

Code and full finite outputs: <https://github.com/Mohammad-Aldebbbeh/prime-offsets>.